Final project: VR Psychological Experiment Room

Fundamentals of eXtended Reality – Master HCI – Université Paris-Saclay

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In this final project done by a group of three, you will develop a virtual environment stimulating a virtual reality psychological experiment room. The project objective is to develop a virtual environment to be used by psychological researchers to study human perception, behaviour, and emotional responses. The virtual room could be modified in terms of its size, lighting, colours, sounds, furniture arrangement, or the presence of virtual objects, allowing researchers to investigate how environmental changes influence people's feelings and behaviour. Users as participants in such a study could be asked to complete simple tasks, make decisions, estimate distances, or report their level of comfort, stress, attention, or presence. The system could record measurable responses such as task completion time, choices, and questionnaire results. This project would provide you an opportunity to explore **VR as a tool for psychological research**, while also introducing concepts such as experimental control, user interaction, data collection, and the design of immersive environments.

Step 1: Create the virtual environment

It is recommended to use SweetHome3D (https://www.sweethome3d.com/) to create a virtual space for a museum or a historical site. You can design the room and its decorations using this software and extract the results (separately) in the fbx format to be imported into Unity engine. You can also use other software (Blender, 3Ds Max, Maya, etc.) to create 3D models as you see fit. You can also find other 3D objects for your scene(s) but be aware of the objects' size.

Step 2: Set up the scene

A set of different tasks is to be designed and developed for the project. Any interactive objects or scenes should have a Reset button to reset their settings so the next user or participant can redo them again. Whatever you choose to develop in this application, it needs to include the following interaction and concepts:

- Virtual navigation by touch pad using gliding/flying metaphor
- Manipulation with ray casting at an offset distance or a virtual hand for close-up manipulation
- Activation / Deactivation of objects during runtime
- Canvas + 3D TextMeshPro
- Physics
- Lighting controlled during the interaction

- 3D audio integrated in the scene
- Timer that counts down the time
- Scripting of object behaviours and logics in the scene

Evaluation

I will evaluate the project based on different criteria: relevance, creativity, complexity, interaction, and user experience. The completeness is also taken into account. It means **it is better to have something simple that works than many functions implemented but none of them works**. It is, therefore, recommended to have several Unity scenes as back up every time you add some drastic changes in the project. The quality of the scripts and explicative comments within them will be looked at as well.

Submission

The final project should be uploaded to a sharable repository before midnight on Thursday October 22, 2026, and its link is sent to me by email (huyen.nguyen@universite-paris-saclay.fr). A **video** recording showing your application should also be included in this repository. *Attention*: The Unity project itself should not be bigger than **2GB**.

A ReadMe.md that lists the members of the group, contribution of each member, and some bullet points describing what you have implemented as well as the mechanism of the application is needed as well. Failure to adhere to these requirements will results in a malus point to your project.